Problem Sheet I: Revision of Groups, Rings and Fields

From the class book: Appendix A (A.1 to A.5) and Appendix B

1. Formula sheet: groups

- Write $n = p_1^{a_1} \cdots p_k^{a_k}$.
- φ is Euler's function, $\tau(n)$ the number of divisors, $p(a)$ the number of partitions of a.

Quantity	Formula / value
Generators of $\mathbb{Z}_n$	$\varphi(n)$
Subgroups of $\mathbb{Z}_n$	$\tau(n)$, exactly one of each order $d \mid n$
Elements of order d in $\mathbb{Z}_n$ ($d \mid n$)	$\varphi(d)$
Order of a^k	$o(a)/\gcd(o(a), k)$
Order of $(a, b) \in G \times H$	$\operatorname{lcm}(o(a), o(b))$
$\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn}$	iff $\gcd(m, n) = 1$
Order of a permutation	lcm of its cycle lengths; an r-cycle is even iff r is odd
Permutations of cycle type $1^{a_1} 2^{a_2} \cdots$ in S_n	$n! / \prod_k k^{a_k} a_k!$
Conjugacy classes of S_n	$p(n)$: 3, 5, 7, 11, 15 for $n = 3, \ldots, 7$
Class size	$\lvert\mathrm{cl}(a)\rvert = [G : C(a)]$; $\lvert G\rvert = \lvert Z(G)\rvert + \sum [G : C(a)]$
$\#\,\mathrm{Hom}(\mathbb{Z}_m, \mathbb{Z}_n)$	$\gcd(m, n)$; onto ones: $\varphi(n)$ if $n \mid m$; one-one: $\varphi(m)$ if $m \mid n$
$\#\,\mathrm{Hom}(\mathbb{Z}_m, G)$	$\#\{x \in G : x^m = e\}$
$\#\,\mathrm{Hom}(G, A)$, A abelian	$\#\,\mathrm{Hom}(G/G', A)$; e.g. $S_n/A_n \cong \mathbb{Z}_2$ for $n \geq 2$
Automorphism groups	$\mathrm{Aut}(\mathbb{Z}_n) \cong U(n)$, $\mathrm{Aut}(\mathbb{Z}) \cong \mathbb{Z}_2$, $\mathrm{Aut}(K_4) \cong S_3$, $\mathrm{Aut}(Q_8) \cong S_4$, $\mathrm{Aut}(\mathbb{Z}_p^n) \cong GL(n, \mathbb{Z}_p)$, $\lvert\mathrm{Aut}(D_n)\rvert = n\varphi(n)$ ($n \geq 3$)
$\lvert GL(n, \mathbb{Z}_p)\rvert$	$(p^n - 1)(p^n - p) \cdots (p^n - p^{n-1})$: $\lvert GL(2, \mathbb{Z}_2)\rvert = 6$, $\lvert GL(3, \mathbb{Z}_2)\rvert = 168$
$U(n)$ cyclic	iff $n = 1, 2, 4, p^k, 2p^k$ (p odd); $U(2^k) \cong \mathbb{Z}_2 \times \mathbb{Z}_{2^{k-2}}$, $k \geq 3$
Sylow	$\lvert G\rvert = p^a m$, $p \nmid m$: $n_p \equiv 1 \pmod p$, $n_p \mid m$; $n_p = 1$ iff the Sylow p-subgroup is normal
Abelian groups of order n	$p(a_1) \cdots p(a_k)$; $p(1), \ldots, p(6) = 1, 2, 3, 5, 7, 11$
Groups of order $n = 1, \ldots, 16$	1, 1, 1, 2, 1, 2, 1, 5, 2, 2, 1, 5, 1, 2, 1, 14

Boxed results (groups)

- $G/Z(G)$ cyclic $\Rightarrow G$ abelian; $\mathbf{Inn}(G) \cong G/Z(G)$.
- Groups of order p^2 are abelian and p-groups have non-trivial centre.
- Every group of order n is cyclic iff $\gcd(n, \varphi(n)) = 1$; in particular $|G| = pq$, $p < q$, $p \nmid q - 1$ forces G cyclic.
- Groups of order pq, p^2q, pqr, p^n ($n \geq 2$) are never simple; A_5 is the smallest non-abelian simple group.

Standard groups at a glance

Group	Order	Subgroups	Normal subgroups	Classes	$\lvert Z(G)\rvert$	Elements of order 2
K_4	4	5	5	4	4	3
S_3	6	6	3	3	1	3
Q_8	8	6	6	5	2	1
D_4	8	10	6	5	2	5
D_5	10	8	3	4	1	5
A_4	12	10	3	4	1	3
S_4	24	30	4	5	1	9
A_5	60	59	2	5	1	15

2. Formula sheet: rings

In $\mathbb{Z}_n$, $n = p_1^{a_1} \cdots p_k^{a_k}$	Count
Units	$\varphi(n)$
Non-zero zero divisors	$n - \varphi(n) - 1$
Idempotents	2^k
Nilpotents	$n/(p_1 \cdots p_k)$
Ideals (= subrings = additive subgroups)	$\tau(n)$
Prime ideals = maximal ideals	k, namely $\langle p_i \rangle$
Ring homomorphisms $\mathbb{Z}_m \to \mathbb{Z}_n$	$2^{\omega(n) - \omega(n/\gcd(m,n))}$, ω = number of distinct primes; a unital one exists iff $n \mid m$

Boxed results (rings)

- In a commutative ring with unity: R/M is a field iff M is maximal; R/P is an integral domain iff P is prime; so maximal $\Rightarrow$ prime.
- $\mathbb{Z}_n$ is a field iff it is an integral domain iff n is prime; a finite integral domain is a field.
- $\mathbf{char}(\mathbb{Z}_m \times \mathbb{Z}_n) = \mathbf{lcm}(m, n)$.
- In $R[x]$: f is a unit iff a_0 is a unit and the other coefficients are nilpotent; f is nilpotent iff all coefficients are nilpotent.
- $R[x]$ is a PID iff R is a field.
- Field $\subset$ ED $\subset$ PID $\subset$ UFD $\subset$ ID, all strict.

Boxed results (quotients)

- $F[x]/\langle f_1^{e_1} \cdots f_r^{e_r} \rangle$ (distinct monic irreducibles f_i) has r maximal ideals and $\prod(e_i + 1)$ ideals.
- Maximal ideals of $\mathbb{Z}[x]$ are $\langle p, f \rangle$ with f irreducible mod p; $\langle x \rangle$, $\langle p \rangle$, $\langle x^2 + 1 \rangle$ are prime, not maximal.
- $|\mathbb{Z}[i]/\langle a + bi \rangle| = a^2 + b^2$; $\mathbb{Z}[i]/\langle n \rangle$ is a field iff n is a prime $\equiv 3 \pmod 4$.
- Maximal ideals of $C[0, 1]$ are $M_c = \{f : f(c) = 0\}$; $M_n(F)$ is simple.

3. Formula sheet: fields

Topic	Fact
Tower law	$[K : F] = [K : E][E : F]$
Cyclotomic	$[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$, $\mathbf{Gal} \cong U(n)$; $[\mathbb{Q}(\cos \frac{2\pi}{n}) : \mathbb{Q}] = \varphi(n)/2$ ($n \geq 3$); $x^n - 1$ has $\tau(n)$ irreducible factors over $\mathbb{Q}$
Splitting field degrees over $\mathbb{Q}$	$x^3 - 2$: 6 (S_3); $x^4 - 2$: 8 (D_4); $x^4 + 1$: 4 ($\mathbb{Z}_2 \times \mathbb{Z}_2$); $x^p - a$: $p(p-1)$
$\mathbb{F}_{p^n}$	exists and is unique for every p^n; $\mathbb{F}_{p^m} \subseteq \mathbb{F}_{p^n}$ iff $m \mid n$; $\tau(n)$ subfields
Multiplicative group	$\mathbb{F}_q^\times$ is cyclic of order $q - 1$; $\varphi(q-1)$ generators (primitive elements)
Galois group	$\mathbf{Gal}(\mathbb{F}_{p^n}/\mathbb{F}_p) \cong \mathbb{Z}_n$, generated by Frobenius $a \mapsto a^p$
Monic irreducibles of degree n over $\mathbb{F}_q$	$N_q(n) = \frac{1}{n}\sum_{d\mid n} \mu(d) q^{n/d}$; $N_q(2) = \frac{q^2-q}{2}$, $N_q(3) = \frac{q^3-q}{3}$; over $\mathbb{F}_2$: 1, 2, 3, 6, 9 for $n = 2, \ldots, 6$

Irreducibility tests

- (1) Degree 2 or 3: irreducible iff no root in the field.
- (2) Rational root test over $\mathbb{Q}$.
- (3) Eisenstein at a prime p, possibly after the shift $x \mapsto x+1$ (e.g. $\Phi_p(x)$).
- (4) Mod p test: if f is monic and irreducible mod some p, it is irreducible over $\mathbb{Q}$ (the converse fails: x^4+1).
- (5) Gauss: a primitive polynomial in $\mathbb{Z}[x]$ irreducible over $\mathbb{Z}$ is irreducible over $\mathbb{Q}$.
- (6) Degree 4 with no root: also rule out a product of two quadratics.

4. Standard counter-examples

False statement	Counter-example
Converse of Lagrange	A_4 has no subgroup of order 6
Abelian $\Rightarrow$ cyclic	K_4, $(\mathbb{Q},+)$
All proper subgroups cyclic $\Rightarrow$ cyclic	K_4, Q_8, S_3
All subgroups normal $\Rightarrow$ abelian	Q_8
Normality is transitive	$\langle (12)(34)\rangle \trianglelefteq V_4 \trianglelefteq A_4$, but not normal in A_4
Same class equation $\Rightarrow$ isomorphic	Q_8 and D_4
Every element of finite order $\Rightarrow$ finite group	$\mathbb{Q}/\mathbb{Z}$, $(\mathcal{P}(\mathbb{N}),\triangle)$
Maximal $\Rightarrow$ prime (ring without unity)	$4\mathbb{Z}$ in $2\mathbb{Z}$
Prime $\Rightarrow$ maximal	$\langle x\rangle$ in $\mathbb{Z}[x]$; $\{0\}$ in $\mathbb{Z}$
Irreducible $\Rightarrow$ prime	2 in $\mathbb{Z}[\sqrt{-5}]$
UFD $\Rightarrow$ PID	$\mathbb{Z}[x]$, $F[x,y]$
PID $\Rightarrow$ ED	$\mathbb{Z}[\frac{1+\sqrt{-19}}{2}]$
No root $\Rightarrow$ irreducible (degree ≥ 4)	x^4+4, $(x^2+1)^2$ over $\mathbb{Q}$
Irreducible over $\mathbb{Q} \Rightarrow$ irreducible mod some p	x^4+1
Algebraic $\Rightarrow$ finite extension	$\overline{\mathbb{Q}}/\mathbb{Q}$

5. How to attack a counting question

1. Identify the object first: is it really $\mathbb{Z}_n$, $U(n)$, S_n, D_n, a product, or a quotient? Rewrite products by CRT ($\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ for coprime m,n) and quotients by the isomorphism theorems.
2. Split into prime parts. For abelian groups and for $\mathbb{Z}_n$, counts are multiplicative over the primes, so handle each p-part and multiply.
3. Elements of a given order: list the cycle types (for S_n) or the component orders (for products) whose lcm is the target, then count each case with the formula.

4. Homomorphisms: a hom out of a cyclic group is fixed by the image of a generator; a hom into an abelian group factors through G/G'.
5. Subgroups: count cyclic subgroups of order d as (elements of order d)$/\varphi(d)$.
6. Rings: reduce to $\mathbb{Z}_{p^a}$ or $F[x]/\langle f^e\rangle$ factors by CRT, then multiply units, idempotents and ideals; nilpotents also multiply.
7. Fields: use the tower law with known degrees; for $\mathbb{F}_q$ everything is governed by divisors of the exponent and of $q-1$.
8. Check the answer on a small case ($n=6$ or 12) before committing in a NAT question; in an MSQ, test each option separately.

> **Exam shortcut.**
> - For a cyclic subgroup count, never add elements of order d and divide by d: divide by $\varphi(d)$.
> - For subgroups of order p in $\mathbb{Z}_p \times \mathbb{Z}_p$ the answer is $(p^2-1)/(p-1) = p+1$.

6. Solved mixed problems

Example 1. How many elements of order 6 are there in S_6?

Solution

- Cycle types in S_6 with lcm 6: a 6-cycle, or a 3-cycle with a disjoint transposition (and one fixed point). 6-cycles: $6!/6 = 120$.
- Type $(3,2,1)$: $6!/(3\cdot 2\cdot 1) = 120$. Total 240.

Example 2. For which $n \in \{15, 21, 33, 35, 39, 255\}$ is every group of order n cyclic?

Solution

- Use $\gcd(n, \varphi(n)) = 1$. $\varphi(15) = 8$, $\varphi(33) = 20$, $\varphi(35) = 24$, $\varphi(255) = \varphi(3)\varphi(5)\varphi(17) = 128$: all coprime to n, so cyclic only.
- $\varphi(21) = 12$ and $\varphi(39) = 24$ share the factor 3 with n (as $3 \mid 7-1$, $3 \mid 13-1$), and non-abelian groups of orders 21 and 39 exist. Answer: 15, 33, 35, 255.

Example 3. Find (a) $\#\,\mathrm{Hom}(\mathbb{Z}_{12}, S_3)$ (b) $\#\,\mathrm{Hom}(\mathbb{Z}_6, S_4)$ (c) $\#\,\mathrm{Hom}(S_4, \mathbb{Z}_6)$.

Solution

- (a) Count $x \in S_3$ with $x^{12} = e$: orders 1, 2, 3 all divide 12, so all 6 elements: 6.
- (b) Count $x \in S_4$ with $x^6 = e$, i.e. order 1, 2, 3 (S_4 has no element of order 6): $1 + 9 + 8 = 18$.
- (c) $\mathbb{Z}_6$ is abelian, so homs factor through $S_4/S_4' = S_4/A_4 \cong \mathbb{Z}_2$: $\#\,\mathrm{Hom}(\mathbb{Z}_2, \mathbb{Z}_6) = \gcd(2,6) = 2$.

Example 4. In $\mathbb{Z}_{25} \times \mathbb{Z}_5$ find the number of elements of order 5 and of subgroups of order 5.

Solution

- Elements with $5x = 0$: 5 choices in $\mathbb{Z}_{25}$ (multiples of 5) times 5 in $\mathbb{Z}_5$, i.e. 25, of which 24 are non-zero, all of order 5.
- Each subgroup of order 5 is cyclic with $\varphi(5) = 4$ generators: $24/4 = 6$ subgroups.

Example 5. For $R = \mathbb{Z}_{200}$ find the number of units, nilpotents, non-zero zero divisors, idempotents, ideals and maximal ideals.

Solution

- $200 = 2^3 \cdot 5^2$, $k = 2$. Units $\varphi(200) = 200 \cdot \frac{1}{2} \cdot \frac{4}{5} = 80$.
- Nilpotents $200/10 = 20$ (multiples of 10).
- Non-zero zero divisors $200 - 80 - 1 = 119$. Idempotents $2^2 = 4$.
- Ideals $\tau(200) = 4 \cdot 3 = 12$.
- Maximal (= prime) ideals: $\langle 2 \rangle, \langle 5 \rangle$, so 2.

Example 6. Let $R = \mathbb{Z}_5[x]/\langle x^4 - 1 \rangle$. Find the number of maximal ideals, ideals, idempotents and units. Is R an integral domain?

Solution

- Over $\mathbb{Z}_5$, $x^4 - 1 = (x-1)(x-2)(x-3)(x-4)$ since $\mathbb{F}_5^\times$ has order 4. By CRT, $R \cong \mathbb{Z}_5^4$. Maximal ideals: 4. Ideals: $2^4 = 16$. Idempotents: $2^4 = 16$. Units: $4^4 = 256$.
- Not an integral domain ($x - 1$ is a zero divisor).

Example 7. (a) Count monic irreducible cubics over $\mathbb{F}_3$ and quartics over $\mathbb{F}_2$. (b) In $\mathbb{F}_{16} = \mathbb{F}_2[x]/\langle x^4 + x + 1\rangle$, find the order of $\bar{x}$, the number of primitive elements and the subfields. (c) Find $[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}]$ and the minimal polynomial of $\sqrt{2} + \sqrt{3}$ over $\mathbb{Q}$.

Solution

- (a) $N_3(3) = (27 - 3)/3 = 8$; $N_2(4) = (16 - 4)/4 = 3$ (namely $x^4 + x + 1$, $x^4 + x^3 + 1$, $x^4 + x^3 + x^2 + x + 1$).
- (b) $\bar{x}$ has order dividing 15; $\bar{x}^3 \neq 1$ and $\bar{x}^5 = \bar{x}^2 + \bar{x} \neq 1$, so the order is 15 and $\bar{x}$ generates $\mathbb{F}_{16}^{\times}$.
- Primitive elements: $\varphi(15) = 8$.
- Subfields $\mathbb{F}_{2^d}$, $d \mid 4$: $\mathbb{F}_2, \mathbb{F}_4, \mathbb{F}_{16}$.
- (Contrast: a root of $x^4 + x^3 + x^2 + x + 1 = \Phi_5$ has order 5, so that quartic is irreducible but not primitive.)
- (c) The degrees 2 and 3 both divide $[\mathbb{Q}(\sqrt{2}, \sqrt[3]{2}) : \mathbb{Q}] \leq 6$, so it is 6.
- For $\alpha = \sqrt{2} + \sqrt{3}$: $\alpha^2 = 5 + 2\sqrt{6}$, $(\alpha^2 - 5)^2 = 24$, so $\alpha^4 - 10\alpha^2 + 1 = 0$; since $\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ has degree 4, $x^4 - 10x^2 + 1$ is the minimal polynomial.
- It is irreducible over $\mathbb{Q}$ yet reducible mod every prime, like $x^4 + 1$.